CivicOrbit (CivicMitra)

Updated Project Deliverables

Design & Implementation Report • Test Case Sheet • Issue / Bug Log • Updated Screenshots

Project	CivicOrbit (CivicMitra) - Gamified Sustainability Platform
Institute	K.K. Wagh Institute of Engineering Education & Research, Nashik
Department	Computer Engineering (B.Tech, Third Year)
Team members	[Add team member names]
Guide	[Add guide name]
Version / Date	v1.1 - 01 October 2026

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Part 1 - Design & Implementation Report

1.1 Introduction

CivicOrbit (earlier CivicMitra) is a smart-city sustainability platform that turns eco-friendly daily habits into a game. Users complete challenges, upload photo proof, earn points and take part in a community. This report documents the changes made in the latest iteration: an improved AI-based proof verification (fraud detection) and a new date feature, along with the updated website.

1.2 Summary of Changes

#	Change	Type	Module
1	Fraud detection fixed - proof images are now checked against the challenge's required proof and rejected with a reason if they do not match	Bug fix	Challenge Verification
2	Date feature added - current date shown in a badge on the landing page	New feature	Landing Page / UI
3	Website updated - refreshed hero section, call-to-action buttons and verification feedback screen	Enhancement	Frontend UI

1.3 Design Changes

Fraud detection flow. Earlier, a submitted proof image could be approved even when it had no relation to the challenge (a fraud loophole). In the updated design, every challenge carries a *Proof Required* description (for example, *'Upload a photo of your bucket setup in the bathroom'*). On submission, the image is analysed by the AI verification service and compared with this requirement. The service returns an approval status and a short human-readable reason. Points are credited only when the status is **Approved**.

Step	Description
1	User opens a challenge and reads the instructions and proof requirement
2	User uploads a proof photo
3	Image + challenge proof requirement are sent to the AI verification service
4	Service decides Approved / Not Approved and returns a reason
5	If Approved: points are awarded. If Not Approved: reason is displayed and the user can use 'Try Again'

Date feature. A pill-shaped badge with a calendar icon and the current date (e.g. *Thursday, 1 October 2026*) is displayed at the top of the landing page, above the headline. The date is generated on the client from the system clock and formatted as weekday, day, month, year.

1.4 Module-wise Implementation Details

Module	Implementation details
Landing Page	Hero section with headline 'Small habits. Big planet.', tagline, date badge, and two actions: 'Start Your Eco Journey' and 'Sign In'. Decorative leaf background in the brand green theme.
Date Feature	Current date read from the system clock, formatted in long form and rendered in a badge component with a calendar icon.
Challenges	Each challenge shows title, instructions and the Proof Required box with an image upload option (example: Bucket Bath Challenge).

Module	Implementation details
Verification (Fraud Detection)	Uploaded image is validated against the challenge proof requirement. Irrelevant images (e.g. a glass of water instead of a bucket setup) are rejected.
Feedback UI	'Verification Not Approved' state with an alert icon, explanation of why the image failed, and a 'Try Again' button.

1.5 Technology Stack

Web front end deployed online (Vercel), AI-based image verification, backend services and database as per the project architecture. Refer to the SRS for the full stack.

1.6 Results

• Irrelevant proof images are now rejected with a clear reason, protecting point integrity.

• Users get immediate feedback and can retry without leaving the challenge.

• The landing page now shows the current date, improving the overall experience.

Part 2 - Test Case Sheet

Test cases cover the modules changed in this iteration. Actual output for TC-01 to TC-05 is based on the captured screenshots (Part 5).

Test ID	Module	Input	Expected Output	Actual Output	Status
TC-01	Date Feature	Open landing page	Badge shows current date in long format	Badge shows 'Thursday, 1 October 2026'	Pass
TC-02	Landing Page	Open landing page	Headline, tagline, 'Start Your Eco Journey' and 'Sign In' buttons visible	All elements displayed correctly	Pass
TC-03	Fraud Detection	Bucket Bath Challenge: upload photo of a glass of water	Verification rejected with reason	'Verification Not Approved - image shows a glass of water being poured, not a bucket setup'	Pass
TC-04	Feedback UI	Rejected verification	'Try Again' button shown	'Try Again' button displayed	Pass
TC-05	Challenge Modal	Open Bucket Bath Challenge	Title, instructions and Proof Required box shown	Displayed correctly	Pass
TC-06	Fraud Detection	Bucket Bath Challenge: upload genuine bucket photo	Verification approved, points awarded	[Fill after running test]	[Pass/Fail]
TC-07	Fraud Detection	Upload unrelated image (e.g. landscape) to any challenge	Verification rejected	[Fill after running test]	[Pass/Fail]
TC-08	Date Feature	Change device date, reload page	Badge reflects the new date	[Fill after running test]	[Pass/Fail]

Part 3 - Issue / Bug Log

Issue ID	Description	Severity	Root Cause	Fix	Status	Date
BUG-01	Fraud detection allowed irrelevant proof images to be accepted for a challenge	High	Proof image was not strictly validated against the challenge's proof requirement	Verification now compares the image with the required proof and rejects mismatches with a reason	Fixed	01-Oct-2026
BUG-02	Rejected users received no clear guidance on what to do next	Medium	No dedicated failure state in the verification dialog	Added 'Verification Not Approved' screen with explanation and 'Try Again' button	Fixed	01-Oct-2026
ENH-01	Current date not shown on the landing page	Low	Feature not implemented	Added date badge with calendar icon above the hero headline	Done	01-Oct-2026

Note: edit root causes and dates to match your team's actual records if they differ.

Part 4 - Updated Source Code

The latest working code is submitted as a ZIP file / GitHub repository link (tagged commit).

Item	Details
Repository link	[Paste GitHub repository URL]
Tagged commit / release	[e.g. v1.1]
Version	v1.1 - 01 October 2026
Version notes	1) Fixed fraud detection in challenge proof verification. 2) Added date feature on landing page. 3) Updated website UI and verification feedback screen.

Part 5 - Updated Screenshots

Screenshots reflecting the current, tested functionality.

5.1 Date Feature - Landing Page

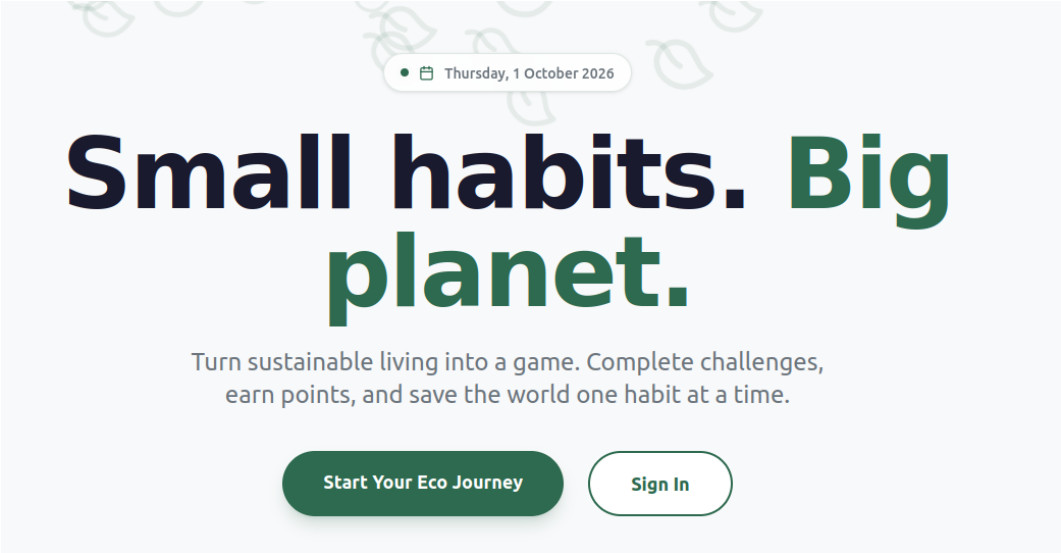


Figure 1: Updated landing page with the date badge ('Thursday, 1 October 2026') above the headline.

5.2 Fraud Detection - Verification Not Approved

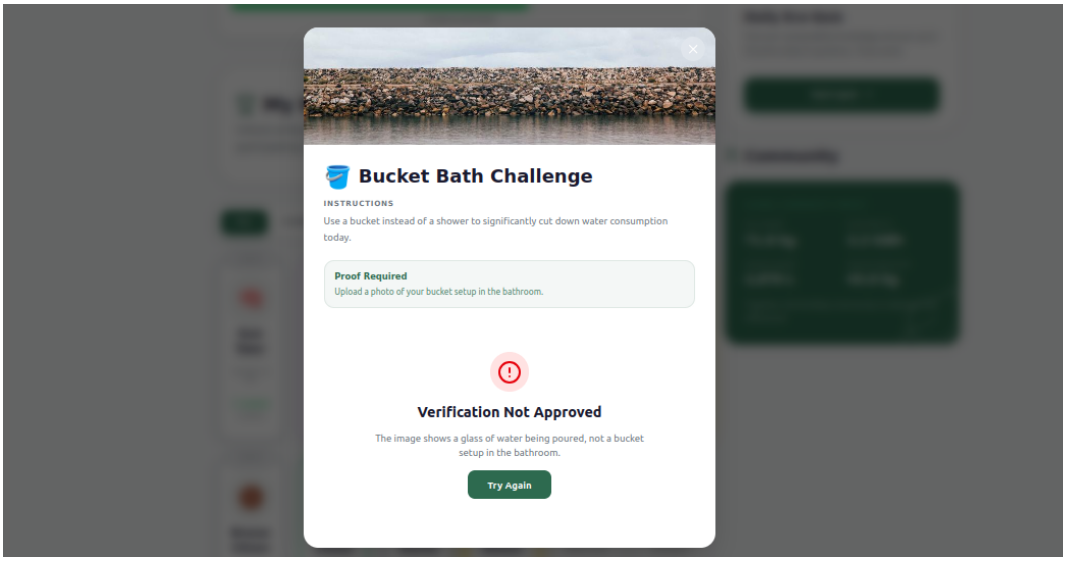


Figure 2: Bucket Bath Challenge - the uploaded image (a glass of water) is rejected because it does not match the required bucket setup.